

Text Mining and Natural Language Processing

Course syllabus. Instructors: Kirill Maslinsky, Sergey Pashakhin.

MA programme “Data Analytics for Politics and Society”, Higher School of Economics — Saint Petersburg, 2021. 36 hours.

Abstract

For social and political sciences, written text provide essential data for studying ideology and political discourse, conflict, sentiment and political affiliation, among many other things. With a growing availability of larger collections of text in digital form it is tempting to scale the research up in terms of the population studied (e.g. “all of twitter”), time spans (e.g. “all of the American history”), and geographical scope (e.g. “all foreign ties of China”). Computational methods for text analysis promise to aid at the scale where traditional content analysis is not feasible.

We will use R programming environment as a toolbox for text analysis. To “learn by doing” we will work with real text collections and will replicate some methods from the recent social research employing computational text analysis.

Course Objective

The goal of the course is to provide basic understanding on how to properly use collections of texts as quantitative evidence, and to make this knowledge practical.

Planned Learning Outcomes

- Understanding possibilities of the automated text analysis as well as its pitfalls and important caveats about applying statistical tests to language data.
- Being able to apply computational methods of text analysis (e.g. analysis of word frequency and co-occurrence, document classification, topic modeling) to collections of texts.
- Being able to adequately interpret and report the results of computational text analysis in research papers.

Course Content

1. Counting words — Preprocessing: transforming text into data in R.
2. Comparing corpora — Comparing word usage in contrast corpora.
3. Document-level modeling — Vector space model. Document classification.
4. Co-occurrence — Distributional semantics.
5. Dictionary methods — Sentiment.

Assessment

Homework

Assessment Criteria Written Assignments (Homework analysis report) «Excellent» (8-10) Has a clear argument, which addresses the topic and responds effectively to all aspects of the task. Fully satisfies all the requirements of the task; rare minor errors occur; «Good» (6-7) Responds to most aspects of the topic with a clear, explicit argument. Covers the requirements of the task; may produce occasional errors. «Satisfactory» (4-5) Generally addresses the task; the format may be inappropriate in places;

display little evidence of (depending on the assignment): independent thought and critical judgement include a partial superficial coverage of the key issues, lack critical analysis, may make frequent errors. «Fail» (0-3) Fails to demonstrate any appropriate knowledge.

Sample Assignments

1. Select any two presidents from the dataset. Build a frequency list and a wordcloud for each of them. Use the list of stopwords from the stopwords package.
2. The same as (1), but use the most frequent words from our dataset as a list of stopwords (experiment with different number of stopwords).
3. (*) Make two Zipf-plots for speeches before and after 1917. Hint: there is an example in the Tidytext book.

Class participation

Assessment Criteria In-class participation/tutorials and paper summaries «Excellent» (8-10) A critical analysis which demonstrates original thinking and shows strong evidence of preparatory research and broad background knowledge. «Good» (6-7) Shows strong evidence of preparatory research and broad background knowledge. Excellent oral expression. «Satisfactory» (4-5) Satisfactory overall, showing a fair knowledge of the topic, a reasonable standard of expression. Some hesitation in answering follow-up questions and/or gives incomplete or partly irrelevant answers. «Fail» (0-3) Limited evidence of relevant knowledge and an attempt to address the topic. Unable to offer relevant information or opinion in answer to follow-up questions

Final project

Assessment Criteria «Excellent» (8-10) A well-structured, analytical presentation of project work. Shows strong evidence and broad background knowledge. In a group presentation all members contribute equally and each contribution builds on the previous one clearly; Answers to follow-up questions reveal a good range and depth of knowledge beyond that covered in the presentation and show confidence in discussion. «Good» (6-7) Clearly organized analysis, showing evidence of a good overall knowledge of the topic. The presenter of the project work highlights key points and responds to follow up questions appropriately. In group presentations there is evidence that the group has met to discuss the topic and is presenting the results of that discussion, in an order previously agreed. «Satisfactory» (4-5) Takes a very basic approach to the topic, using broadly appropriate material but lacking focus. The presentation of project work is largely unstructured, and some points are irrelevant to the topic. Knowledge of the topic is limited and there may be evidence of basic misunderstanding. In a group presentation, most of the work is done by one or two students and the individual contributions do not add up. «Fail» (0-3) Fails to demonstrate any appropriate knowledge.

Sample Assignments

 Sample project topics

- A system for recommendation of relevant emoji to chat participants
- Analysis of political agenda of State Duma of Russian Federation based on session transcripts
- Language usage on the service “Yandex.District” in Saint-Petersburg

Grading Formula

$0.2 \text{class participation} + 0.5 \text{homework} + 0.3 \text{final project}$

References

Core Reading

- Aggarwal, C. C., & Zhai, C. (Eds.). (2012). Mining text data. Springer Science & Business Media.
- Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political analysis*, 21(3), 267-297.
- Rule, A., Cointet, J. P., & Bearman, P. S. (2015). Lexical shifts, substantive changes, and continuity in State of the Union discourse, 1790–2014. *Proceedings of the National Academy of Sciences*, 112(35), 10837-10844.

Additional Reading

- Zachary C Lipton (2018). “The Mythos of Model Interpretability.” In: *Queue* 16.3, p. 30
- Justin Grimmer (2010). “A Bayesian hierarchical topic model for political texts: Measuring expressed agendas in Senate press releases.” In: *Political Analysis* 18.1, pp. 1–35
- William L Hamilton et al. (2016). “Inducing domain-specific sentiment lexicons from unlabeled corpora.” In: *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. Conference on Empirical Methods in Natural Language Processing. Vol. 2016. NIH Public Access, p. 595

Software

- Python 3+